Chapter 10. Causal Discovery

Constraint-based, score-based and functional methods; local causal discovery

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10.1 Causal discovery

Motivation. How much of the causal diagram can be recovered from data, under which assumptions and algorithms?

10.1.1 The structure-learning problem

Motivation. Input, output and assumptions of causal discovery.

Causal discovery goes from $P(\mathbf{V})$ (or samples) to $\mathcal{G}$. Does the rooster's crow cause sunrise? Binary (rooster, sun) observations are almost always equal, but data alone do not give the arrow's direction.

Structure learning. *Input:* data i.i.d. from $P(\mathbf{V})$. *Output:* the graphs consistent with the data (hopefully including $\mathcal{G}$); an unsupervised problem. Extensions: multiple data sets, interventional data, unknown interventions.

10.1.1.1 DAG models

A DAG model has one node per variable and an edge per direct functional relationship; it corresponds to a Markovian model (*causal sufficiency*: no unmeasured confounder). Three equivalent definitions, for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$:

- **Factorization:** $P(A,B,C,D,E) = P(E \mid D)P(D \mid B,C)P(C \mid A)P(B \mid A)P(A)$.
- **Local Markov property:** $(C \perp\!\!\!\perp B \mid A)$, $(D \perp\!\!\!\perp A \mid B,C)$, $(E \perp\!\!\!\perp A,B,C \mid D)$.
- **Global Markov property:** $\mathbf{A}$ d-separated from $\mathbf{B}$ by $\mathbf{C}$ in $\mathcal{G} \Rightarrow \mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

10.1.1.2 Faithfulness

The global Markov property is one-directional, and a complete graph is compatible with every $P(\mathbf{V})$; minimality needs the converse.

Definition (Faithfulness)

For all $\mathbf{A},\mathbf{B},\mathbf{C}$: $\mathbf{A}$ is d-separated from $\mathbf{B}$ given $\mathbf{C}$ in $\mathcal{G}$ *iff* $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

Unfaithful distributions have measure zero, but nearly unfaithful ones are many and hard to distinguish in finite samples, and nature may evolve unfaithful configurations (homeostasis, equilibria). Justify faithfulness for the problem.

Example 10.1 (Violation by cancellation)

$X = \epsilon_X$, $Z = X + \epsilon_Z$, $Y = X - Z + \epsilon_Y$ (edges $X \xrightarrow{1} Z$, $Z \xrightarrow{-1} Y$, $X \xrightarrow{1} Y$). X, Y are d-connected, yet $Y = -\epsilon_Z + \epsilon_Y$, so $X \perp\!\!\!\perp Y$: the paths cancel.

10.1.2 Constraint-based structure learning

Motivation. Find the CIs that hold and keep the graphs consistent with them.

10.1.2.1 What constraints a DAG encodes

For $X \to W \to Y$, $X \to Z \leftarrow Y$, of

$$(1)\ X \perp\!\!\!\perp Y \mid W, \quad (2)\ X \perp\!\!\!\perp Y, \quad (3)\ X \perp\!\!\!\perp Y \mid W, Z, \quad (4)\ X \perp\!\!\!\perp Y \mid Z,$$

only (1) is forced; (2)–(4) are dependences ($X \to W \to Y$ open; collider Z opened). Under faithfulness CIs refine the possible DAGs; with no CI information the best guess is the *complete undirected* graph.

Adjacency. If $A \perp\!\!\!\perp B \mid \mathbf{S}$ for some $\mathbf{S} \subseteq \mathbf{V} \setminus \{A, B\}$, then A, B are not adjacent (adjacent variables are dependent given every set). Here only X–Y is removed (separator $\{W\}$), giving the *skeleton* $X - W - Y$, $X - Z - Y$.

Orientation. For an *unshielded triple* $A - C - B$ with separator $\mathbf{S}$ of A, B: if $C \notin \mathbf{S}$, orient the *unshielded collider* (v-structure) $A \to C \leftarrow B$; otherwise C is a non-collider. Here $Z \notin \{W\}$ gives $X \to Z \leftarrow Y$, ruling out $X \to Z \to Y$, $X \leftarrow Z \leftarrow Y$, $X \leftarrow Z \to Y$; $X - W - Y$ is a non-collider of undetermined direction.

10.1.2.2 Markov equivalence

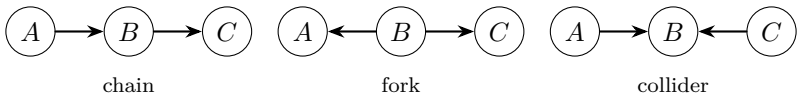


Figure. *Chain and fork both imply $A \perp\!\!\!\perp C \mid B$ (Markov equivalent); the collider implies $A \perp\!\!\!\perp C$, $A \not\perp\!\!\!\perp C \mid B$, so only colliders are orientable.*

$X \to W \to Y$, $X \leftarrow W \leftarrow Y$, $X \leftarrow W \to Y$ (each with $X \to Z \leftarrow Y$) entail the same CIs: they are *observationally* (Markov) *equivalent*. A *Markov equivalence class* (MEC) is a set of such DAGs.

Theorem 10.1 (Verma and Pearl 1990)

Two DAGs represent the same statistical model iff they have the same skeleton and the same unshielded colliders.

Proof [Supplement]. *If.* Whether a path is open given $\mathbf{S}$ depends only on which inner nodes are colliders and their relation to $\mathbf{S}$. Unshielded triples agree by assumption; a shielded triple can be bypassed by the shielding edge, so a shortest open path has no differing shielded colliders. Hence the d-separations, and so the Markov models, coincide. *Only if.* Different skeletons: an edge $A - B$ in one DAG only; in the other A, B are separated by some set (e.g. parents of the later one), never in the first. Same skeleton, but $A \to C \leftarrow B$ unshielded collider in one only: in the first some separator of A, B excludes C; in the second every separator contains C ($A - C - B$ is open otherwise). Either way the d-separations differ. ■

A class is represented by one mixed graph, the *pattern* or CPDAG (completed partially directed acyclic graph). On three variables the 25 DAGs form 11 classes (Eberhardt 2017): chain, reversed chain and fork on X, Y, Z share one; the collider $X \to Y \leftarrow Z$ is alone.

10.1.2.3 The PC algorithm

Basic idea:

(1) non-adjacent vertices have a separator;

(2) for an unshielded collider $X \to Z \leftarrow Y$, the separator excludes Z.

Meek's rules.

Algorithm 1 PC algorithm (Spirtes and Glymour)

start with the complete undirected graph $\mathcal{G}$ on n vertices
for $i = 0, \ldots, n-2$ **do** ▷ skeleton
  for every adjacent pair A, B in $\mathcal{G}$ **do**
    if $\exists \mathbf{S}$ with $|\mathbf{S}| = i$, $\mathbf{S} \subseteq \mathrm{adj}_{\mathcal{G}}(A) \cup \mathrm{adj}_{\mathcal{G}}(B) \setminus \{A, B\}$, and $A \perp\!\!\!\perp B \mid \mathbf{S}$ **then**
      remove $A - B$; sepset$[\{A, B\}] \leftarrow \mathbf{S}$
for every non-adjacent A, B with $A - C - B$ **do** ▷ v-structures
  if $C \notin$ sepset$[\{A, B\}]$ **then** orient $A \to C \leftarrow B$
orient as many remaining edges as possible without creating new v-structures or directed cycles (Meek's rules)

1. $b - c \Rightarrow b \to c$ if $a \to b$ with a, c non-adjacent (else new v-structure $a \to b \leftarrow c$).
2. $a - b \Rightarrow a \to b$ if $a \to c \to b$ (else a cycle).
3. $a - b \Rightarrow a \to b$ if $a - c \to b$ and $a - d \to b$ with c, d non-adjacent (if $b \to a$, Rule 2 forces $c \to a$, $d \to a$: new v-structure).
4. $a - b \Rightarrow a \to b$ if $a - c \to d$ and $c \to d \to b$ with c, b non-adjacent and a, d adjacent.

Example 10.2 (Walk-through)

True DAG $A \to B$, $A \to C$, $B \to D$, $C \to D$; CIs $B \perp\!\!\!\perp C \mid A$ and $A \perp\!\!\!\perp D \mid \{B, C\}$.

- $|\mathbf{S}| = 0$: all pairs dependent.
- $|\mathbf{S}| = 1$: remove $B - C$, sepset$[\{B, C\}] = \{A\}$.
- $|\mathbf{S}| = 2$: remove $A - D$, sepset$[\{A, D\}] = \{B, C\}$. Skeleton $A - B$, $A - C$, $B - D$, $C - D$.
- V-structures: $D \notin \{A\}$ gives $B \to D \leftarrow C$; $B - A - C$, $A - B - D$, $A - C - D$ are non-colliders.
- Meek's rules: one might apply Rule 1 to $A - B$ (via $B \to D$) and $A - C$, giving $A \to B$, $A \to C$, $B \to D \leftarrow C$, the true DAG.

Caution: Rule 1 needs an arrow *into* B; $B \to D$ points out. No rule fires, and the CPDAG is $A - B$, $A - C$, $B \to D \leftarrow C$, whose class is $\{A \to B, A \to C\}$, $B \to A \to C$, $C \to A \to B$ (excluding $B \to A \leftarrow C$, a new v-structure). The true DAG is one member.

Exercise. Apply PC to the five-variable DAG above. $B \to D \leftarrow C$ is found, and Rule 1 orients $D \to E$ ($B \to D$, B, E non-adjacent).

10.1.2.4 Extensions

Conservative PC (Ramsey, Spirtes and Zhang 2006). Split faithfulness into *adjacency faithfulness* (adjacent $\Rightarrow$ no separator) and *orientation faithfulness*: for an unshielded collider $X \to Z \leftarrow Y$, X, Y are dependent given every subset of $\mathbf{V} \setminus \{X, Y\}$ containing Z; otherwise every separator contains Z. Check all subsets of X's and Y's potential parents: Z in no separator gives a v-structure, in all a non-collider, otherwise the triple is marked "unfaithful" (underlined). Robust to orientation-faithfulness violations at modest cost.

Order-independent PC (Colombo and Maathuis 2014). An early false-negative CI test changes later adjacency sets, so PC's output depends on variable order (skeletons vary over 25 random orderings). Remedies: average over orderings, or remove edges only after all tests of one conditioning-set size; orientation uses a majority rule over separating sets (conservative is too restrictive).

Fast Causal Inference (FCI) allows latent confounders. True graph $X \to Y$, $W \to Z$, latent

$U \to Y$, $U \to Z$: the skeleton is $X - Y - Z - W$ and FCI outputs $X \circ\!\!\to Y \leftrightarrow Z \leftarrow\!\!\circ W$, flagging a confounder of Y, Z. PAG edge marks:

$X_1 \quad X_2$ (no edge)	X_1 and X_2 are not adjacent
$X_1 \circ\!\!\to X_2$	X_2 is not an ancestor of X_1
$X_1 \circ\!\!-\!\!\circ X_2$	no set d-separates X_1 and X_2
$X_1 \to X_2$	X_1 is a cause of X_2
$X_1 \leftrightarrow X_2$	there is a latent common cause of X_1 and X_2

10.1.2.5 Summary

With a CI oracle PC is complete (recovers the CPDAG). CI tests: partial correlation (linear Gaussian), Fisher's exact or χ^2 (categorical), kernel- or model-based (Appendix 10.1.8). Relaxing sufficiency or faithfulness is active research.

10.1.3 Score-based approaches

Motivation. Search for the best-fitting graph under a complexity penalty.

10.1.3.1 Why scores?

A single wrong CI test can propagate through skeleton and orientation. Score-based methods score fit and simplicity and search over graphs; hybrids (e.g. MMHC) combine both paradigms.

Why it is possible. Relative to the truth (e.g. $E \to A \leftarrow B$, $A \to S$), an extra arc adds parameters and wrong causal claims; a missing arc cannot be compensated by fitting parameters.

Intuition. Truth $A \to B \leftarrow C$ ($A \perp\!\!\!\perp C$, $A \not\perp\!\!\!\perp C \mid B$); proposal $A \to B \to C$ claims the opposite, so it fits the data poorly, as measured by the log-likelihood.

10.1.3.2 Log-likelihood for DAG models

With $P(\mathbf{V}) = \prod_i P(V_i \mid Pa_{\mathcal{G}}(V_i))$ and models P_{θ_i}, for data $D = \{\mathbf{v}^{(j)}\}_{j=1}^n$,

$$\ell(\theta; D) = \sum_{j=1}^{n} \sum_{i=1}^{k} \log P_{\theta_i}(v_i^{(j)} \mid \mathbf{pa}_{\mathcal{G}}^{(j)}(V_i)),$$

a *decomposable* likelihood: each θ_i is fitted separately.

For the true $A \to B \leftarrow C$ and proposed $B \to A$, $C \to B$:

$$\ell_{\text{left}} = \textstyle\sum_j \left[\log P_A(a_j) + \log P_B(b_j \mid a_j, c_j) + \log P_C(c_j)\right],$$
$$\ell_{\text{right}} = \textstyle\sum_j \left[\log P_A(a_j \mid b_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)\right].$$

With saturated local models, $\ell_{\text{left}}/n - \ell_{\text{right}}/n = I(A; C \mid B) > 0$ (Appendix 10.1.7).

The problem. Against the complete $B \to A$, $C \to A$, $C \to B$, $\ell_{\text{right}} = \sum_j [\log P_A(a_j \mid b_j, c_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)]$ fits *better*: $\{P : A \perp\!\!\!\perp C\}$ is a submodel of all $P(A, B, C)$. Likelihood alone always picks a complete graph (overfitting); parameters must be penalized.

10.1.3.3 BIC

$$\text{BIC} = -2\,\ell(\hat{\theta}; D) + m \log n,$$

m the number of parameters; lower is better. For a linear Gaussian network $X_i = \beta_{0i} + \sum_{X_k \in Pa(X_i)} \beta_{ki} X_k + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$, node i has $|Pa(X_i)| + 2$ parameters, so $m = |E| + 2|V|$.

10.1.3.4 Search and GES

DAGs are super-exponentially many, so exhaustive scoring is impossible; local search moves between states by local moves able to reach any DAG. **Greedy Equivalence Search** (GES; Chickering 2002) uses equivalence classes as states:

1. Start with the empty graph.
2. *Addition phase:* move to the best class reachable by adding one edge, until no improvement. (The result can depend on the representative: adding $B - C$ to $\{A \to B \;\; C,\; A \leftarrow B \;\; C\}$ gives $\{A \to B \leftarrow C\}$ or the chain/fork class.)
3. *Removal phase:* likewise, removing one edge.
4. Return the current class.

Theorem 10.2 (Chickering 2002)

Under faithfulness and as $n \to \infty$, GES identifies the true equivalence class.

Proof [Supplement]. With faithfulness and a consistent score such as BIC, as $n \to \infty$:

(a) adding an edge that creates a dependence present in P, or removing one whose dependence is absent, increases the score (local consistency);

(b) the score is equal for Markov-equivalent DAGs. By (a) and Meek's conjecture (proved by Chickering), the forward phase stops at a class whose DAGs are I-maps of the true independence model; by (a) the backward phase stops only at a minimal I-map, which under faithfulness is the true class. Details: Chickering (2002), Theorem 4. ■

Operators (on CPDAGs). Insert$(u, v, \mathbf{T})$: for non-adjacent u, v add $u \to v$, with $\mathbf{T} \subseteq \mathrm{Ne}(v) \setminus \mathrm{Adj}(u)$; orient $t \to v$ for $t \in \mathbf{T}$; convert to a CPDAG. Delete$(u, v, \mathbf{H})$: remove the u–v edge, with $\mathbf{H} \subseteq \mathrm{Ne}(v) \cap \mathrm{Adj}(u)$; orient $h \to v$, $h \to u$ for $h \in \mathbf{H}$; convert to a CPDAG. GES applies the operator with the best score gain.

Example 10.3

Truth $A \to B \leftarrow C$. Forward: Insert$(A, B, \emptyset)$ gives $A - B$; Insert$(C, B, \{A\})$ ($A \in \mathrm{Ne}(B) \setminus \mathrm{Adj}(C)$) gives $A \to B \leftarrow C$; no further gain. Backward: no deletion helps. Output $A \to B \leftarrow C$.

10.1.3.5 Continuous optimization: NOTEARS and DAGMA

NOTEARS (Zheng et al. 2018) fits a linear SEM (noise not necessarily Gaussian) by black-box optimization. For $X = \epsilon_X$, $Z = aX + \epsilon_Z$, $Y = bX + cZ + \epsilon_Y$,

$$\begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{pmatrix} \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} + \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix} \iff \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b + ac & c & 1 \end{pmatrix} \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix}.$$

With data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $W \in \mathbb{R}^{d \times d}$,

$$\min_{W \in \mathbb{R}^{d \times d}} \frac{1}{2n} \|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda \|W\|_1 \quad \text{subject to} \quad h(W) = \mathrm{tr}\left(e^{W \circ W}\right) - d = 0.$$

For a binary adjacency matrix B, B^k counts length-k walks, so $\mathrm{tr}(e^B) = \sum_k \mathrm{tr}(B^k)/k!$ counts weighted closed walks including the d trivial ones: $\mathrm{tr}(e^B) = d$ iff B is a DAG. $\mathrm{tr}(e^{W \circ W})$ is a smooth

weighted analogue, solved by augmented Lagrangian (Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning).

NOTEARS-MLP (Zheng et al. 2020) learns general f: $\min_f L(f)$ s.t. $\mathcal{G}(f) \in$ DAGs, $L(f) = \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, f_j(\mathbf{X}))$, $\mathbb{E}[X_j \mid X_{\mathrm{pa}(j)}] = g_j(f_j(X))$, $\mathbb{E}f_j(X) = 0$, with $k \to j$ iff f_j depends on X_k. Since f_j is free of X_k iff $\|\partial_k f_j\|_{L^2} = 0$, set $[W(f)]_{kj} = \|\partial_k f_j\|_{L^2}$ and solve $\min_{f_j \in H^1(\mathbb{R}^d)} L(f)$ s.t. $h(W(f)) = 0$ (H^1: Sobolev space). For $\mathrm{MLP}(u; A^{(1)}, \ldots, A^{(h)}) = \sigma(A^{(h)} \cdots \sigma(A^{(2)}\sigma(A^{(1)}u)))$, the output is free of u_k if column k of $A^{(1)}$ is zero, giving

$$\min_{\theta} \ \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, \mathrm{MLP}(\mathbf{X}; \theta_j)) + \lambda \|A_j^{(1)}\|_{1,1} \quad \text{s.t.} \quad h(W(\theta)) = 0,$$

with

$$[W(\theta)]_{kj} = \|k\text{-th column of } A_j^{(1)}\|_2.$$

DAGMA (Bello, Aragam and Ravikumar 2022). Matrix exponentials are costly and hard to optimize on large graphs; DAGMA uses the log-determinant penalty

$$h_{\mathrm{DAGMA}}(W) = -\log\det(sI - W \circ W) + d\log s, \qquad s > 0 \ (\text{e.g. } s = 1),$$

zero for acyclic graphs and growing fast near highly cyclic regions. It optimizes $\min_\theta Q(f_\theta; \mathbf{X}) + \beta_1\|\theta\|_1 + h_{\mathrm{DAGMA}}(W(\theta))$ for linear or nonlinear SEMs, faster with competitive accuracy; an optimization device, not a new assumption.

10.1.3.6 Summaries

Score-based. BIC/GES guarantees are large-sample only; search is intractable if the score is not polynomial-time; misspecified local models $P(V_i \mid Pa_{\mathcal{G}}(V_i))$ give wrong graphs; no hidden variables.

Constraint- vs. score-based. Constraint-based (PC, FCI): CI tests, sepsets and Meek's rules; output CPDAG (PAG for FCI); sensitive to CI errors. Score-based (GES, NOTEARS, DAGMA): optimize a score over graphs; better in small samples, needs correct specification. Both assume faithfulness and (except FCI) sufficiency; hybrids (MMHC) combine them.

10.1.4 Parametric approaches

Motivation. Functional-form assumptions orient edges that CIs cannot.

With two variables there are no CIs; $X \to Y$ versus $Y \to X$ needs asymmetries in the joint distribution.

Additive noise model (ANM). $y = f_y(x) + u_y$, $x \perp\!\!\!\perp u_y$. Compare

$$\text{H1: } y = f_y(x) + u_y,\ x \perp\!\!\!\perp u_y\ (x \to y) \qquad \text{vs.} \qquad \text{H2: } x = f_x(y) + u_x,\ y \perp\!\!\!\perp u_x\ (y \to x).$$

Two main cases:

(1) f non-linear;

(2) f linear, u_y non-Gaussian.

10.1.4.1 Non-linear functions

Generically an ANM holds in one direction only; in the other the residual's spread depends on the regressor.

1. Fit $\hat{y} = \hat{f}_y(x)$ and $\hat{x} = \hat{f}_x(y)$ by any ML method.

2. Residuals $\hat{u}_x = x - \hat{f}_x(y)$, $\hat{u}_y = y - \hat{f}_y(x)$.
3. Test independence (e.g. HSIC): $\hat{u}_y \perp\!\!\!\perp x$ suggests $X \to Y$; $\hat{u}_x \perp\!\!\!\perp y$ suggests $Y \to X$; neither casts doubt on the ANM.

10.1.4.2 Linear functions with non-Gaussian noise: LiNGAM

Cramér's decomposition theorem: if a sum of two independent real random variables is Gaussian, both are; by induction, for any finite sum.

LiNGAM (linear non-Gaussian acyclic model). For $y = ax + u_y$ with non-Gaussian errors,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \qquad x = u_x, \quad y = au_x + u_y.$$

Regressing x on y leaves a mixture of u_x, u_y uncorrelated with but (Darmois–Skitovich/Cramér) not independent of y; forward, the residual u_y is independent of x.

The trick: ICA. Independent non-Gaussian sources $s_1, \ldots, s_n$ are observed as $\mathbf{x} = A\mathbf{s}$. Mixtures look more Gaussian than sources, so ICA finds directions maximizing non-Gaussianity, measured by:

- *Kurtosis* $\kappa(y) = \frac{\mathbb{E}[(y-\mu)^4]}{\mathrm{Var}(y)^2} - 3$: zero for Gaussians, positive for heavy tails (Laplace), negative for light tails (uniform); outlier-sensitive.
- *Negentropy* $J(y) = H(y_{\text{Gauss}}) - H(y) \geq 0$ (y_{Gauss} Gaussian, same mean and variance), zero iff y is Gaussian.

ICA returns W_{ICA} with $\hat{\mathbf{s}} = W_{\text{ICA}}\mathbf{x}$ up to permutation and scaling (Hyvärinen and Oja 2000). For two variables, run ICA on (x, y) and find the zero in the mixing matrix.

LiNGAM (Shimizu et al. 2006). $x_i = \sum_{j<i} b_{ij}x_j + e_i$, i.e. $\mathbf{x} = B\mathbf{x} + \mathbf{e}$, $\mathbf{x} = (I - B)^{-1}\mathbf{e}$; let $W = I - B$. ICA gives $W_{\text{ICA}} = PDW$ (P permutation, D diagonal), resolved by the structure:

1. $\mathbf{e} = W\mathbf{x}$, W a row-permuted, rescaled $I - B$.
2. $W_{\text{ICA}} = \text{ICA}(\mathbf{X})$.
3. Permute rows so the diagonal has no zeros (unique, as $I - B$ has non-zero diagonal and is permutable to lower-triangular), rescale to unit diagonal: $W = \text{normalize}(\text{rowpermute}(W_{\text{ICA}}))$.
4. $\hat{B} = I - W$; the causal order makes $\hat{B}$ lower-triangular.

Summary. In an ANM $y = f(x) + u_Y$ with f non-linear or U_Y non-Gaussian, and no latent confounding ($X \perp\!\!\!\perp U_Y$), the direction is identifiable.

10.1.4.3 Score matching (NoGAM)

(Montagna et al. 2023.) For $X_i = f_i(\text{PA}_i) + N_i$ with arbitrary noise, the *score* $s(X) = \nabla_X \log p(X)$, $s_i(X) = \partial_{X_i} \log p(X)$ (unrelated to BIC scores), is estimable by score matching without $p(X)$. It identifies *leaves*; removing them recursively gives a topological order, then extra edges are pruned.

Why leaves are special. With $\log p(X) = \sum_k \log p_k(N_k)$, $N_k = X_k - f_k(\text{PA}_k)$,

$$s_l(X) = \partial_{N_l} \log p_l(N_l) \quad \text{for a leaf } X_l,$$

$$s_i(X) = \partial_{N_i} \log p_i(N_i) - \sum_{k \in \text{CH}_i} \partial_{N_k} \log p_k(N_k)\, \partial_{X_i} f_k \quad \text{otherwise.}$$

For a leaf the regression residual is $R_l = X_l - q(X_{-l}) = N_l$, so s_l is a function of R_l; for non-leaves the child terms remain. Under ANM identifiability,

$$X_i \text{ is a leaf} \iff \min_h \mathbb{E}\big[\{h(R_i) - s_i(X)\}^2\big] = 0.$$

Algorithm (NoGAM). (1) Estimate $\hat{s}(X) \approx \nabla_X \log p(X)$.

(2) For each remaining node, regress X_i on X_{-i}; residuals $\hat{R}_i$.

(3) Predict $\hat{s}_i(X)$ from $\hat{R}_i$; record the MSE.

(4) The smallest-MSE node is a leaf; remove it and repeat.

(5) Reverse the removal order and prune extra edges; no Gaussian noise assumed.

10.1.5 Appendix: GES operators in detail

Insert$(u, v, \mathbf{T})$ adds $u \to v$ ($u \notin \text{Adj}(v)$), orients $t \to v$ for $t \in \mathbf{T}$ ($t \notin \text{Adj}(u)$), and completes to a CPDAG. Requires $\mathbf{T} \subseteq \text{Ne}(v) \setminus \text{Adj}(u)$, $(\text{Ne}(v) \cap \text{Adj}(u)) \cup \mathbf{T}$ a clique, and every semi-directed path from v to u through this set. In the v-structure example, after $A - B$ the move is Insert$(C, B, \{A\})$.

Delete$(u, v, \mathbf{H})$ removes $u - v$ or $u \to v$, orients $h \to v$ and $h \to u$ for $h \in \mathbf{H}$, and completes to a CPDAG. Requires $\mathbf{H} \subseteq \text{Ne}(v) \cap \text{Adj}(u)$ and $(\text{Ne}(v) \cap \text{Adj}(u)) \setminus \mathbf{H}$ a clique. These are Chickering's (2002) operators; XGES reparametrises them (Nazaret et al. 2025).

10.1.6 Appendix: Post-nonlinear causal model (optional)

PNL: for $X_1 \to X_2$, $X_2 = g_2\{f_1(X_1) + E_2\}$, $E_2 \perp\!\!\!\perp X_1$, g_2 invertible, so $E_2 = g_2^{-1}(X_2) - f_1(X_1)$. Find ψ_2, ϕ_1 with $\psi_2(X_2) - \phi_1(X_1)$ independent of X_1; if this works in one direction only, orient the edge. Identifiable generically, with special ambiguous cases (Zhang and Hyvärinen 2009).

10.1.7 Appendix: Scores in detail

Why the left likelihood is larger. For left $A \to B \leftarrow C$, right $B \to A$, $C \to B$, saturated maximized population log-likelihoods are

$$\ell_{\text{left}}/n = -H(A) - H(C) - H(B \mid A, C), \qquad \ell_{\text{right}}/n = -H(C) - H(B \mid C) - H(A \mid B).$$

If the collider is true, $A \perp\!\!\!\perp C$ and $H(A) + H(C) + H(B \mid A, C) = H(A, B, C)$, so

$$\ell_{\text{left}}/n - \ell_{\text{right}}/n = H(A \mid B) + H(B \mid C) - H(A) - H(B \mid A, C) = I(A; C \mid B),$$

positive exactly when $A \not\perp\!\!\!\perp C \mid B$.

Bayesian Dirichlet scores (categorical). BD (Cooper and Herskovits) is the posterior probability of a network; the user must set every α_{ijk}, and equivalent structures may score differently. BDe is score-equivalent via an equivalent sample size α; BDeu sets $\alpha_{ijk} = \alpha/(r_i q_i)$ (uniform over CPT cells, not over graphs). With i variables, j parent configurations (q_i), k values (r_i), Dirichlet conjugate to multinomial,

$$\text{BD}(\mathcal{G}, D; \alpha) = \prod_{i=1}^{N} \text{BD}(X_i \mid Pa_{X_i}; \alpha_i) = \prod_{i=1}^{N} \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + n_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + n_{ijk})}{\Gamma(\alpha_{ijk})},$$

α_i the equivalent sample size. $\alpha_{ijk} = 1$: K2; $\alpha_{ijk} = 1/2$: Jeffreys; $\alpha_{ijk} = \alpha/(r_i q_i)$: BDeu ($\alpha_i = \alpha$).

BIC (maximization form). Dividing by -2,

$$\text{BIC}(\mathcal{G}, D) = \sum_{i=1}^{N} \Big[\ell_i(\hat{\theta}_i) - \frac{\log n}{2} q_i(r_i - 1)\Big], \qquad \ell_i(\hat{\theta}_i) = \sum_{m=1}^{n} \log \hat{P}_i(x_i^{(m)} \mid \pi_i^{(m)}),$$

score-equivalent; it is the Laplace approximation of log BDeu, agreeing up to $O(1)$.

10.1.8 Appendix: CI tests for causal discovery

χ^2 test (categorical). Tests $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$ for all z. Example (2008 General Social Survey, main news source by age; credit Yibi Huang):

Age	TV	Internet	Newspapers	Other	Total
18–29	109	92	25	36	262
30–49	272	157	88	63	580
50+	345	59	165	63	632
Total	726	308	278	162	1474

Expected counts under CI are $E(x, y \mid z) = N(x, z)N(y, z)/N(z)$, and

$$\chi^2(Z = z) = \sum_{x,y} \frac{(O(x, y \mid z) - E(x, y \mid z))^2}{E(x, y \mid z)}, \qquad \chi^2_{\text{total}} = \sum_z \chi^2(Z = z),$$

compared with χ^2_{df}, $df = (|X| - 1)(|Y| - 1)|Z|$.

Partial correlation (linear).

$$r_{XY \cdot Z} = \frac{r_{XY} - r_{XZ}r_{YZ}}{\sqrt{1 - r^2_{XZ}}\sqrt{1 - r^2_{YZ}}}$$

for scalar Z (multivariate: recursion or residuals). A drop means Z explains the correlation.

HSIC (non-linear marginal independence). Tests $H_0 : X \perp\!\!\!\perp Y$. With Gram matrices K, L and $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, $\widehat{\text{HSIC}}(X, Y) = \frac{1}{n^2}\operatorname{tr}(KHLH)$; with characteristic kernels, HSIC $= 0$ iff $X \perp\!\!\!\perp Y$. Null by permutation, gamma approximation or asymptotics (Gretton et al. 2005); for conditional independence use KCI.

Kernel conditional independence test (KCI). Tests $H_0 : X \perp\!\!\!\perp Y \mid Z$: with kernels K_X, K_Y, K_Z, regress out Z via $R_Z = \epsilon(K_Z + \epsilon I)^{-1}$, $\tilde{K}_X = R_Z K_X R_Z$, $\tilde{K}_Y = R_Z K_Y R_Z$, and test $T_{\text{KCI}} = \frac{1}{n}\operatorname{tr}(\tilde{K}_X \tilde{K}_Y) = 0$. Replaces Fisher-$z$/$\chi^2$ in PC for non-linear CI, at higher cost and with bandwidth sensitivity (Zhang, Peters, Janzing and Schölkopf 2011).

10.2 Local causal discovery

Motivation. What changes when structure learning reasons over local objects (blankets, neighborhoods, order prefixes)?

10.2.1 The map

Motivation. Organize discovery algorithms by signal and reach.

The global workhorses were PC, FCI (CI tests) and GES, NOTEARS, DAGMA (one global score). Here we go local, along two knobs:

- **Signal**: test independences or optimize a score?
- **Reach**: whole graph or a local object?

	global	local
CI tests	PC, FCI	GS, HITON
CI → score		MMHC
score	GES, DAGMA	BOSS

This lecture fills the local column.

10.2.1.1 One primitive: grow–shrink

- **Grow:** add Y while $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$ (CI) or while the score improves.
- **Shrink:** drop Y if $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$ (CI) or if the score improves.

Algorithm	Local object	Signal
GS	Markov blanket	CI
HITON	one target's parents and children	CI
MMHC	skeleton, then DAG	CI + score
BOSS	order prefix	score

10.2.2 Going local on CI

Motivation. Learn Markov blankets and target neighborhoods by CI tests.

10.2.2.1 The Markov blanket

$$\mathrm{MB}(X) = \big(\mathrm{pa}(X) \cup \mathrm{ch}(X) \cup \mathrm{pa}(\mathrm{ch}(X))\big) \setminus \{X\},$$

parents, children and spouses. Given $\mathrm{MB}(X)$, all other variables are irrelevant for X; it is small yet drives skeleton and orientation (Margaritis and Thrun 1999; Margaritis 2003).

10.2.2.2 GS: grow–shrink the blanket

Grow (permissive): add any Y with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$. **Shrink**: drop Y once $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$. In $X \to C \to G$, grow may add G; once $C \in \mathbf{S}$, $X \perp\!\!\!\perp G \mid C$ removes it. Then find true neighbors and orient colliders as in PC.

Algorithm 2 GS-MB(X)

1: $\mathbf{S} \leftarrow \emptyset$
2: **repeat** find $Y \in \mathbf{V} \setminus (\{X\} \cup \mathbf{S})$ with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \cup \{Y\}$
3: **until** no such Y exists
4: **repeat** find $Y \in \mathbf{S}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \setminus \{Y\}$
5: **until** no such Y exists
6: **return S**

Correctness sketch. Under faithfulness a parent or child of X is dependent on X given any set, so grow adds it; a spouse becomes dependent once the common child is in $\mathbf{S}$. So $\mathrm{MB}(X) \subseteq \mathbf{S}$ after grow. For $W \in \mathbf{S} \setminus \mathrm{MB}(X)$, $\mathrm{MB}(X) \subseteq \mathbf{S} \setminus \{W\}$ separates X from W, so shrink removes W.

Structure induction. (1) Compute $\mathrm{MB}(X)$ for every X.

(2) $Y \in \mathrm{MB}(X)$ is not a neighbor if $X \perp\!\!\!\perp Y \mid \mathbf{S}$ for some $\mathbf{S} \subseteq \mathbf{R}$, $\mathbf{R}$ the smaller of $\mathrm{MB}(X) \setminus \{Y\}$, $\mathrm{MB}(Y) \setminus \{X\}$.

(3) Orient unshielded colliders by local tests.

(4) Repair cycles and propagate forced directions.

Randomized GS. With large blankets, sample conditioning sets from the smaller blanket and pool tests as evidence: anytime-style, controlled runtime, weaker exactness.

10.2.2.3 HITON: target-focused, not whole-graph

HITON wants only one target T's neighborhood (e.g. feature selection for a disease). HITON-PC($\cdot$) returns a node's *parents and children* (not the PC algorithm) by interleaved grow-while-shrink with small, sample-efficient conditioning sets, run on T and a bounded ring around it. Two pieces: HITON-PC and HITON-MB (Aliferis et al. 2003; 2010).

HITON-MB: two rings, then prune. Example: $P \to T \to C \leftarrow Z$, $C \to G$.

1. Ring 1: PC(T) $\leftarrow$ HITON-PC(T), i.e. P, C.
2. Ring 2: HITON-PC(Y) for each $Y \in$ PC(T); the spouse Z, not adjacent to T, surfaces only here.
3. Prune ring-2 candidates separable from T: G (by C) goes; Z (dependent once C is conditioned on) stays.

Algorithm 3 HITON-MB(T)

PC(T) $\leftarrow$ HITON-PC(T); $\mathbf{C} \leftarrow$ PC(T)
for each $Y \in$ PC(T) **do** $\mathbf{C} \leftarrow \mathbf{C} \cup$ HITON-PC(Y)
for each candidate $X \in \mathbf{C} \setminus$ PC(T) **do**
    if $\exists \mathbf{S} \subseteq \mathbf{C} \setminus \{X\}$ with $T \perp\!\!\!\perp X \mid \mathbf{S}$ **then** remove X from $\mathbf{C}$
return C

10.2.3 Bringing in the score

Motivation. Combine local CI skeletons with scores, and search over orders.

10.2.3.1 MMHC: a hybrid bridge

(Tsamardinos, Brown and Aliferis 2006.) MMPC, the max-min sibling of HITON-PC, finds each node's parents and children; their union is a candidate skeleton, within which a score hill-climb picks a DAG:

$$\text{local PC per node (MMPC)} \;\to\; \text{union} = \text{skeleton} \;\to\; \text{score hill-climb} \;\to\; \text{one DAG}.$$

Sound skeleton restriction, heuristic score-based orientation.

10.2.3.2 BOSS: best order score search

(Andrews et al. 2023.) The state is a permutation π; each v's candidate parents are its *prefix*

$$\text{pre}_\pi(v) = \{w : \pi.\text{index}(w) < \pi.\text{index}(v)\};$$

e.g. for $\pi = (A, C, B, D, E)$, $\text{pre}_\pi(B) = \{A, C\}$.

Scoring an order. The projection $\mathcal{G}_\pi$ picks parents from the prefix by greedy grow/shrink, approximating

$$\text{pa}_\pi(v) \approx \underset{\mathbf{W} \subseteq \text{pre}_\pi(v)}{\arg\max}\ \text{score}_v(\mathbf{W}), \qquad \text{score}_v(\mathbf{W}) = \text{BIC}(X_v \mid X_{\mathbf{W}}),$$

(higher is better), and $\text{score}(\pi) = \sum_v \text{score}_v(\text{pa}_\pi(v))$. By $\text{BIC}(\mathcal{G}, D) = \sum_{v \in \mathbf{V}} \text{BIC}(X_v \mid X_{\text{pa}_\mathcal{G}(v)})$, changing v's parents changes one term; grow–shrink trees (GSTs) cache repeated prefix scores, the key to BOSS's scaling.

Algorithm 4 project$(\mathbf{X}, \pi)$ and best-move$(\mathcal{T}, \pi, v)$

$E \leftarrow \emptyset$
for each v in π **do**
  $\mathbf{Z} \leftarrow \text{pre}_\pi(v)$; $\mathbf{W} \leftarrow \text{grow}(\mathbf{X}, v, \mathbf{Z})$; $\mathbf{W} \leftarrow \text{shrink}(\mathbf{X}, v, \mathbf{W})$
  add $w \to v$ to E for each $w \in \mathbf{W}$
return $\mathcal{G}_\pi = (\mathbf{V}, E)$

$best \leftarrow \mathcal{T}.\text{score}(\pi)$
for $i = 1, \ldots, |\pi|$ **do**
  $\pi' \leftarrow \pi.\text{move}(v, i)$
  if $\mathcal{T}.\text{score}(\pi') > best$ **then** $best \leftarrow \mathcal{T}.\text{score}(\pi')$; $\pi \leftarrow \pi'$
return π

Example 10.4 (Best-move)

Truth $A \to C \leftarrow B$. With $\pi = (A, C, B)$, C is a candidate parent of B, and conditioning on the collider couples A, B: the projection has 3 edges. $\pi' = (A, B, C)$ gives the true collider with 2 edges, the same independences with fewer parameters, so $\text{score}(\mathcal{G}_{\pi'}) > \text{score}(\mathcal{G}_\pi)$ and π' is kept.

Benchmarks. For large, dense models BOSS matches GRaSP's accuracy with far better runtime, compares well with fast GES, PC, DAGMA and LiNGAM, and has run on resting-state fMRI with hundreds to 1000 variables.

10.2.4 Zoom out

Motivation. Choose among the methods and remember what none of them solve.

What to use when.

- Sparse graph, reliable CI tests: GS (full graph) or HITON (one target).
- High-dimensional feature selection: HITON's target blanket.
- Large, dense, score-friendly data: BOSS (or MMHC).
- Hidden confounders: FCI-type methods.
- One effect only: effect-specific local discovery (Entner et al.; LDECC).

Limitations.

- Causal sufficiency: GS, HITON, MMHC, BOSS assume no hidden confounders.
- Faithfulness (or score consistency).
- Finite samples: weak effects and large conditioning sets hurt all.
- Specification and tuning: the local likelihood and penalty set the precision–recall balance.

Locality changes the cost of search, not the assumptions.

Comparison.

	GS	HITON	MMHC	GES	BOSS
Main signal	CI tests	CI tests	CI + score	BIC-like score	BIC-like score
Search object	blankets	target blanket	skeleton + DAG	CPDAGs	orders
Locality	Markov blankets	PC/co-parents	local skeleton	edge operators	prefixes
Output target	MEC graph	variables for T	one DAG	CPDAG	CPDAG
Main risk	CI error	target scope	heuristic orient.	many score calls	model spec.